

BENGAL COLLEGE OF ENGINEERING & TECHNOLOGY



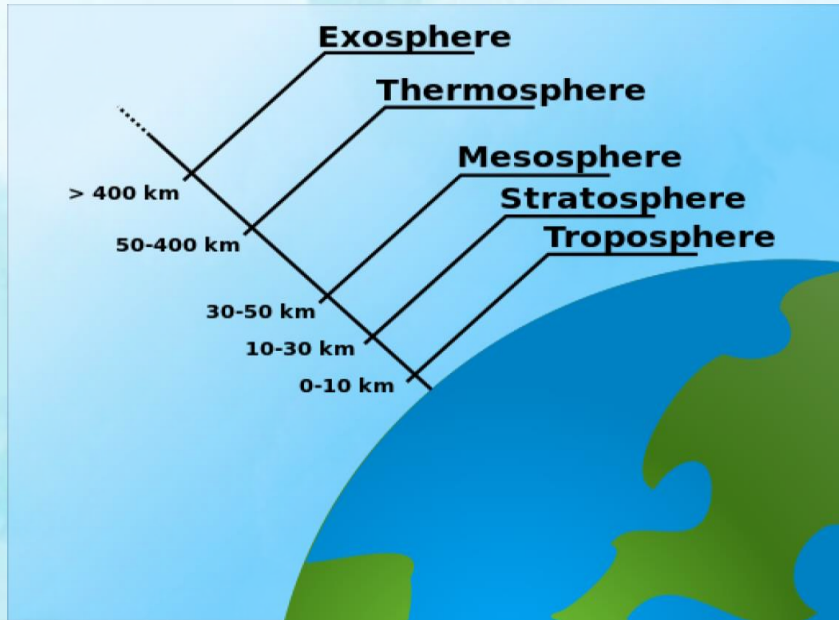
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TOPIC

STRUCTURE OF EARTH ATMOSPHERE



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INTRODUCTION

- The Earth's atmosphere is a vital layer of gases surrounding our planet, essential for life as we know it.
- It protects us from harmful radiation, regulates temperature, and drives weather patterns.
- Understanding its structure is crucial for comprehending climate change and other environmental issues.



METHODOLOGY

- This presentation will utilize existing scientific literature, including data from satellite observations, atmospheric modeling, and meteorological studies.
- Diagrams and visualizations will be used to illustrate the different layers of the atmosphere and their characteristics.
- Key concepts like temperature gradients, chemical composition, and atmospheric pressure will be explained.





DISCUSSION



- The atmosphere is divided into layers based on temperature profiles:
- Troposphere:** The lowest layer, where weather occurs. Temperature decreases with altitude.
- Stratosphere:** Contains the ozone layer, which absorbs UV radiation. Temperature increases with altitude.
- Mesosphere:** Temperature decreases with altitude. Meteors burn up in this layer.
- Thermosphere:** Temperature increases significantly with altitude due to solar radiation. Contains the ionosphere.
- Exosphere:** The outermost layer, gradually merging with space.
- The composition of the atmosphere is primarily nitrogen and oxygen, with trace gases like argon, carbon dioxide, and water vapor.
- Atmospheric pressure decreases with altitude



CONCLUSION

- The layered structure of the Earth's atmosphere plays a crucial role in maintaining life and regulating the planet's climate.
- Understanding the characteristics of each layer is essential for studying atmospheric phenomena and addressing environmental challenges.
- Human activities are impacting the atmosphere's composition and structure, leading to climate change and other environmental problems.





APPLICATIONS

- **Knowledge of atmospheric structure is vital for: Weather forecasting: Understanding atmospheric dynamics is crucial for predicting weather patterns.**
- **Climate modeling: Accurate representation of atmospheric layers is essential for climate models.**
- **Aviation: Aircraft fly in the stratosphere to minimize turbulence.**
- **Space exploration: Understanding the thermosphere and exosphere is crucial for space missions.**
- **Environmental monitoring: Studying changes in atmospheric composition helps monitor pollution and climate change.**



REFERENCE

NASA's Atmospheric Science Data Center:

- Provides detailed information on the different layers of the atmosphere and their significance.

National Geographic:

- Offers an overview of the atmosphere, including its composition and importance for life on Earth.

World Meteorological Organization (WMO):

- Provides insights into atmospheric observations and their role in weather and climate prediction.



**THANK
YOU!!!**

BY- BISHAL ROY